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Constraints Handling Techniques - Stacking Algorithmic Procedure

Constraints Handling Approach for Metaheuristic Algorithms

The optimization models are formulated as constrained minimization problems, expressed as $\min_{\mathbf{w} \in \Omega} f(\mathbf{w})$, where f is a non-convex objective function. Due to the non-convexity of f , metaheuristic optimization methods are employed to explore the search space effectively. These methods are particularly well-suited for navigating complex landscapes characterized by non-convexity and high dimensionality, providing a viable strategy for identifying optimal solutions. The feasible regions, Ω , within the proposed models are classified into two distinct types:

- **Linear Equality Constraint (Ω_1):** The region Ω_1 is defined as:

$$\Omega_1 := \{\mathbf{x} \in \mathbb{R}^S : \mathbf{1}_S^\top \mathbf{x} = 1\},$$

where $\mathbf{1}_S$ denotes a vector of ones of dimension S . This feasible region imposes a linear equality constraint, ensuring that the sum of the ensemble weights equals one.

- **Probability Simplex (Ω_2):** The second region Ω_2 extends Ω_1 by additionally requiring that all weights be non-negative. Inspired by the non-short selling constraints frequently encountered in portfolio optimization, Ω_2 is defined as:

$$\Omega_2 := \{\mathbf{x} \in \mathbb{R}^S : \mathbf{x} \geq \mathbf{0}_S, \mathbf{1}_S^\top \mathbf{x} = 1\}.$$

Given the constrained nature of the optimization problems under consideration, metaheuristic algorithms may generate candidate solutions that lie outside feasible regions. However, for all cases examined, the objective function f is continuously differentiable, and the feasible regions Ω_1 and Ω_2 are closed, convex, non-empty subsets of the function's domain, $\text{dom}(f) \subseteq \mathbb{R}^S$. These properties collectively ensure the existence and uniqueness of orthogonal projections onto the feasible sets Ω_1 and Ω_2 . Consequently, this manuscript proposes the corrective measure of projecting candidate solutions generated outside the feasible regions onto their respective feasible sets using the Euclidean distance metric.

Let $\mathbf{y}_k^i$ denote the i -th individual in the population at iteration k , generated by the stochastic update operator $\mathcal{M}(\cdot)$. This operator, which may include mechanisms such as mutation, crossover, or selection, generates solutions that may not satisfy the constraints of Ω_1 or Ω_2 . In such cases, the corrected solution $\mathbf{w}_k^i$ is obtained by projecting $\mathbf{y}_k^i$ onto the feasible set, denoted generically as:

$$\mathbf{w}_k^i = \mathcal{P}_\Omega(\mathbf{y}_k^i),$$

where the projection operator $\mathcal{P}_\Omega$ is defined as:

$$\mathcal{P}_\Omega(\mathbf{y}) = \arg \min_{\mathbf{z} \in \Omega} \left\{ \frac{1}{2} \|\mathbf{z} - \mathbf{y}\|_2^2 \right\}.$$

Projection onto Ω_1

The projection onto Ω_1 is computed by solving the following optimization problem:

$$\begin{aligned} \min_{\mathbf{z} \in \mathbb{R}^S} \quad & \frac{1}{2} \|\mathbf{z} - \mathbf{y}\|_2^2, \\ \text{s.t.} \quad & \mathbf{1}_S^\top \mathbf{z} = 1. \end{aligned}$$

The Lagrangian for this problem is given by:

$$\mathcal{L}(\mathbf{z}, \mu) = \frac{1}{2} (\mathbf{z} - \mathbf{y})^\top (\mathbf{z} - \mathbf{y}) + \mu (\mathbf{1}_S^\top \mathbf{z} - 1),$$

where $\mu \in \mathbb{R}$ is the Lagrange multiplier. Setting the gradient of $\mathcal{L}$ with respect to $\mathbf{z}$ to zero yields:

$$\nabla_{\mathbf{z}} \mathcal{L} = (\mathbf{z} - \mathbf{y}) + \mu \mathbf{1}_S = 0 \implies \mathbf{z} = \mathbf{y} - \mu \mathbf{1}_S.$$

By substituting $\mathbf{z}$ into the constraint $\mathbf{1}_S^\top \mathbf{z} = 1$, the multiplier μ is obtained as:

$$\mathbf{1}_S^\top \mathbf{y} - \underbrace{\mu \mathbf{1}_S^\top \mathbf{1}_S}_{S\mu} = 1 \implies \mu = \frac{\mathbf{1}_S^\top \mathbf{y} - 1}{S}.$$

The resulting projection onto Ω_1 is thus given by:

$$\mathbf{z} = \mathbf{y} - \left(\frac{\mathbf{1}_S^\top \mathbf{y} - 1}{S} \right) \mathbf{1}_S.$$

In the context of the proposed framework, this yields:

$$\mathbf{w}_k^i = \mathcal{P}_{\Omega_1}(\mathbf{y}_k^i) = \mathbf{y}_k^i - \left(\frac{\mathbf{1}_S^\top \mathbf{y}_k^i - 1}{S} \right) \mathbf{1}_S.$$

Projection onto Ω_2

The projection onto the probability simplex Ω_2 is computed by solving the following constrained optimization problem:

$$\begin{aligned} \min_{\mathbf{z} \in \mathbb{R}^S} \quad & \|\mathbf{z} - \mathbf{y}\|_2^2, \\ \text{s.t.} \quad & \mathbf{1}_S^\top \mathbf{z} = 1, \quad \mathbf{z} \geq \mathbf{0}_S. \end{aligned}$$

This problem entails determining the vector $\mathbf{z}$ on the probability simplex $\Omega_2 := \{\mathbf{x} \in \mathbb{R}^S : \mathbf{x} \geq \mathbf{0}_S, \mathbf{1}_S^\top \mathbf{x} = 1\}$ that minimizes the Euclidean distance to a given vector $\mathbf{y} \in \mathbb{R}^S$. This optimization problem can be formulated as a quadratic program, and given the strict convexity of the objective function, a unique solution is guaranteed. To efficiently compute this solution, Algorithm 1, originally proposed in [1], is implemented. This algorithm has a computational complexity of $\mathcal{O}(S \log S)$ for finding the optimal $\mathbf{z}$, as demonstrated by the authors [1].

Algorithm 1 Euclidean projection of a vector onto the probability simplex

- 1: **Input:** $\mathbf{y} \in \mathbb{R}^S$
 - 2: Sort $\mathbf{y}$ into $\mathbf{u}$: $u_1 \geq u_2 \geq \dots \geq u_S$
 - 3: Find $\rho = \max \left\{ 1 \leq j \leq S : u_j + \frac{1}{j} \left(1 - \sum_{i=1}^j u_i \right) > 0 \right\}$
 - 4: Define $\lambda = \frac{1}{\rho} \left(1 - \sum_{i=1}^\rho u_i \right)$
 - 5: **Output:** $\mathbf{z}$ s.t. $z_i = \max\{y_i + \lambda, 0\}, i = 1, \dots, S$
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Figure 1: Algorithmic procedure for the projection onto Ω_2

Therefore, within the context of the proposed optimization framework, the projection of the i -th individual at the k -th iteration onto the set Ω_2 is expressed as:

$$\mathbf{w}_k^i = \mathcal{P}_{\Omega_2}(\mathbf{y}_k^i) = \max\{\mathbf{y}_k^i + \lambda \mathbf{1}_S, \mathbf{0}_S\},$$

where λ is determined following the procedure outlined in Algorithm 1, and the max operator is applied element-wise.

Stacking Algorithmic Procedure

Algorithm 2 Nested Cross-Validation

Require: Dataset $\mathcal{D}$, Hyperparameter space $\mathcal{H}$, K_{out} , K_{in}

Ensure: Optimal model and hyperparameters

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1: for  $k \leftarrow 1$  to  $K_{out}$  do
2:   Partition  $\mathcal{D}$  into  $\mathcal{D}_{train}^k, \mathcal{D}_{test}^k$ 
3:    $\mathcal{S} \leftarrow \emptyset$ 
4:   for all  $\lambda \in \mathcal{H}$  do
5:      $K_{in}$ -fold CV on  $\mathcal{D}_{train}^k$  with  $\lambda$ 
6:     Compute  $\mu_\lambda$ 
7:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{(\lambda, \mu_\lambda)\}$ 
8:   end for
9:    $\lambda^* = \arg \min_{\lambda \in \mathcal{H}} \mu_\lambda$ 
10:  Train base models on  $\mathcal{D}_{train}^k$ :  $\{f_1, f_2, \dots, f_S\} \leftarrow \text{fit}(\mathcal{D}_{train}^k)$ 
11:  Get estimations from base learners:  $\{\mathbf{o}_1, \dots, \mathbf{o}_S\} \leftarrow \text{predict}(\mathcal{D}_{train}^k, f_1, \dots, f_S), \mathbf{O} = (\mathbf{o}_1, \dots, \mathbf{o}_S)$ .
12:  Compute  $\mathbf{w}$ :  $\mathbf{w} \leftarrow \text{computeWeights}(\mathcal{D}_{train}^k, \mathbf{O}, \lambda^*)$ 
13:  Evaluate  $\{f_1, \dots, f_S, \mathbf{w}\}$  on  $\mathcal{D}_{test}^k$ 
14: end for
15: Select best model
16: return Best model,  $\lambda^*$ 
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Bibliography

- [1] Weiran Wang and Miguel A Carreira-Perpinán. Projection onto the probability simplex: An efficient algorithm with a simple proof, and an application. *arXiv preprint arXiv:1309.1541*, 2013.